

Programming Assignment #7: Online Shopping Cart Part II

1 Overview

More people are taking advantage of curbside pickup offered by grocery stores. For this assignment, continue the previous assignment of writing a C program that simulates an online shopping cart.

- Extend the `ItemToPurchase` struct to contain a new member.
 - `char itemDescription[]` set to "none" in `MakeItemBlank()`
- Implement the `PrintItemDescription()` functions for the `ItemToPurchase` struct which has an `ItemToPurchase` parameter.
- Create 2 additional files for submission.
 - `ShoppingCart.h`
 - `ShoppingCart.c`
- Build the `ShoppingCart` struc with the following data members and related functions
 - Data members
 - * `char customerName[ ]`
 - * `char currentDate[ ]`
 - * `ItemToPurchase cartItems[ ]` with a maximum of 10 slots (can hold up to 10 items of any quantity)
 - * `int cartSize` the number of filled slots in array `cartItems[ ]` (number of items in the cart of any quantity)
 - Related functions
 - * `AddItem()`
 - Adds and item to `cartItems` and has paratmeters of type `ItemToPurchase` and `ShoppingCart`
 - Returns `ShoppingCart` object
 - * `RemoveItem()`

- Removes item from `CartItem`s array (does not set the quantity to 0; removed item should not take up a spot in the array)
- Has a `char [ ]` (an item's name) and a `ShoppingCart` parameter. Returns `ShoppingCart` object
- If item cannot be found, output a message: `Item not found in cart. Nothing removed.`

* `ModifyItem()`

- Modifies an item's description, price, and/or quantity.
- Has parameters of type `ItemToPurchase` and `ShoppingCart`. Returns `ShoppingCart` object.

* `GetNumItemsInCart()`

- Returns the quantity of all items in the cart.
- Has a `ShoppingCart` parameter.

* `GetCostOfCart()`

- Determines and returns the total cost of items in the cart.
- Has `ShoppingCart` parameter.

* `PrintTotal()`

- Outputs total of objects in cart.
- Has a `ShoppingCart` parameter.
- If cart is empty, output a message: `SHOPPING CART IS EMPTY.`
- Outputs total cost of items in cart.

* `PrintDescriptions()`

- Outputs each item's description.
- Has a `ShoppingCart` parameter.
- If cart is empty, output a message: `SHOPPING CART IS EMPTY.`

- In `main()`, prompt the user for a customer's name and today's date. Output the name and date. Create an object of type `ShoppingCart`.
- Implement the `PrintMenu()` function in `main.c`
 - Prints the following menu of options to manipulate the shopping cart.

```
MENU
a - Add item to cart
r - Remove item from cart
c - Change item quantity
i - Output items' descriptions
o - Output shopping cart
q - Quit
```

- Implement the `ExecuteMenu()` function in `main.c`
- In `main()`, call `PrintMenu()` and prompt for the user's choice of menu options. Each option is represented by a single character. If an invalid character is entered, continue to prompt for a valid choice. Once a valid option is entered, execute the option by calling `ExecuteMenu()`. Then print the menu and prompt for a new option. This process should continue until the user enters 'q'.
- Implement each of the menu options in `ExecuteMenu()`.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from `header.c`). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Programming Assignment #7 - Joe Student\n");
```

Your code should meet all the stylistic requirements we have discussed in class including the appropriate use of whitespace and comments.

3 Sample Runs

Your output should match the sample runs below. Sample runs of several menu options are provided.

The figure below shows a sample run of the beginning of the programming and adding an item to the cart.

```
Enter Customer's Name:
Robert Simpson
Enter Today's Date:
December 5, 2024

Customer Name: Robert Simpson
Today's Date: December 5, 2024

MENU
a - Add item to cart
r - Remove item from cart
c - Change item quantity
i - Output items' descriptions
o - Output shopping cart
q - Quit

Choose an option:
a
ADD ITEM TO CART
Enter the item name:
Nike Romaleos
Enter the item description:
Volt color, Weightlifting shoes
Enter the item price:
189
Enter the item quantity:
2
```

The figure below shows the output of menu option i.

```
Choose an option:
i
OUTPUT ITEMS' DESCRIPTIONS
Robert Simpson's Shopping Cart - December 5, 2024

Item Descriptions
Nike Romaleos: Volt color, Weightlifting shoes
Chocolate Chips: Semi-sweet
Powerbeats 2 Headphones: Bluetooth headphones
```

The figure below shows the shopping cart.

```
Choose an option:
o
OUTPUT SHOPPING CART
Robert Simpson's Shopping Cart - December 5, 2024
Number of Items: 8

Nike Romaleos 2 @ $189 = $378
Chocolate Chips 5 @ $3 = $15
Powerbeats 2 Headphones 1 @ $128 = $128

Total: $521

MENU
a - Add item to cart
r - Remove item from cart
c - Change item quantity
i - Output items' descriptions
o - Output shopping cart
q - Quit

Choose an option:
q

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

For this assignment, you must turn in a zip file containing `ItemToPurchase.h`, `ItemToPurchase.c`, `ShoppingCart.h`, `ShoppingCart.c` and `main.c`. Use the built-in zip program in Windows to combine all three files into a single .zip file. Turn in your assignment by attaching the zip file to the assignment submission page. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Due date: The assignment is due by 11:59 pm Monday, December 8th.